

SynthLineage: Submodular-Guided Corpus Distillation and Knowledge Graph Provenance for Collapse-Resistant LLM Fine-Tuning

Vikash Kumar
vikash.kumar4@arm.com

Abstract—When LLMs are repeatedly trained on their own outputs, the resulting model collapse erases rare-event representation and pushes generations toward generic text, a phenomenon now well-characterised theoretically but without practical corpus-level tooling to prevent it. We present SynthLineage, a framework that combines provenance tracking, a Data Lineage Knowledge Graph (DataLineageKG), and a greedy corpus distillation algorithm called *Marginal Maximum Diversity* (MMD). The key theoretical result is that the number of distinct n -grams a corpus covers is monotone submodular, giving greedy coverage-maximizing selection a $(1 - 1/e) \approx 63.2\%$ guarantee; the normalized diversity ratio we report is not itself submodular. Empirically, greedy selection matches or exceeds the best of 1000 random K -subsets on distinct-2. We also introduce the *Ancestral Fidelity Score* $\text{AFS}(s) = \cos(\mathbf{v}_s, \mathbf{v}_{h(s)}) \cdot e^{-\lambda d}$, the first empirical measure of information decay across synthetic generative hops ($\hat{\lambda}=0.569$, $r=0.96$). DistilGPT-2 fine-tuning experiments over three random seeds show that collapse looks like distribution divergence, not reduced surface diversity: the contaminated model generates the most varied text yet scores worst on perplexity and nearly eliminates rare tokens. MMD-Select cuts perplexity by 39.3% ($2.66 \pm 0.09 \rightarrow 1.62 \pm 0.01$) and delivers a $6.7\times$ increase in rare token preservation ($0.35 \pm 0.10\% \rightarrow 2.39 \pm 0.31\%$), while naive depth-based filtering reduces generated-text diversity by 48.4%. A Pearson correlation of $r=+0.999$ between collapse risk and vocabulary size shows that vocabulary growth, the most common corpus health metric, moves in the wrong direction as collapse worsens.

Index Terms—model collapse, submodular optimization, data provenance, knowledge graphs, LLM fine-tuning, rare token preservation, information reuse, data lineage, ancestral fidelity score

I. INTRODUCTION

Training pipelines that crawl the web now inevitably ingest AI-generated content from prior model generations. Shumailov et al. [1] showed what follows: rare patterns disappear first, outputs drift toward bland central tendencies, and each training round drifts further from the human distribution—a process called *model collapse*. Dohmatob et al. [3] derived scaling law bounds on this rate. Gerstgrasser et al. [2] showed that mixing in real data prevents collapse, but this still requires knowing which samples are synthetic and says nothing about which ones to keep.

The selection question is unsolved: given a pool of synthetic samples at various generative depths, which ones should you keep to best preserve the human distribution? Watermarking [4] can flag AI text, but only if the generator is modified before deployment. Simple depth filtering, the obvious

alternative, reduces generated-text diversity (distinct-2) by 48.4% in our experiments. Statistical detectors [14] assess individual samples but ignore corpus-level diversity entirely. SynthLineage addresses this with four contributions:

- 1) Proof that the distinct- n -gram *coverage* of a corpus is monotone submodular (Proposition 1), giving greedy coverage-maximizing selection a $(1 - 1/e)$ bound; the normalized ratio $D-n$ is not itself submodular, and greedy selection empirically matches or exceeds the best of 1000 random K -subsets.
- 2) The Ancestral Fidelity Score, which measures how much of the original human signal a synthetic sample retains; we fit $\hat{\lambda}=0.569$ ($r=0.96$) as the decay constant.
- 3) Diversity Debt, an accounting framework that tracks cumulative diversity loss and triggers human sample injection when needed.
- 4) The first demonstration that collapse manifests as distribution divergence in a *corpus selection* context—Shumailov et al. [1] show rare token loss theoretically; we quantify it as a function of corpus assembly strategy, with MMD-Select recovering a $6.7\times$ improvement in rare token preservation.

Fig. 1 shows the full pipeline.

II. RELATED WORK

A. Model Collapse in Generative AI

Shumailov et al. [1] provided the first comprehensive characterisation of model collapse across LLMs, VAEs, and diffusion models, demonstrating irreversible rare-event erasure under recursive synthetic training. Dohmatob et al. [3] derived scaling law bounds showing collapse rate is governed by the synthetic-to-real data ratio per training round—a result our risk score directly captures. Gerstgrasser et al. [2] proved accumulation prevents collapse if real-data proportion is maintained; this is the key insight we operationalise with the Debt Repayment Rule. Critically, none of these works provide a mechanism for selecting *which* synthetic samples to retain—the problem SynthLineage solves.

B. LLM Training Data Curation and Quality

Large-scale corpora such as C4 [10] and The Pile [11] use deduplication and heuristic scoring to remove low-quality web content, but these filters target human-authored data. AI-generated content is grammatically correct, topically coherent,

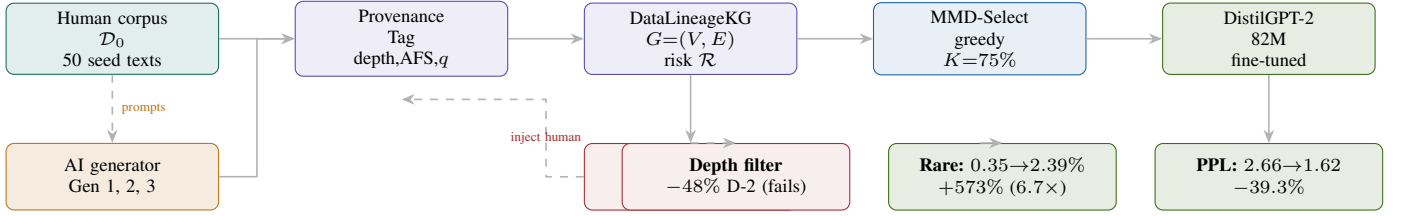


Fig. 1. SynthLineage system architecture. Human corpus $\mathcal{D}_0$ and AI-generated samples are tagged at creation time (ProvenanceTag), assembled into the DataLineageKG, gated by risk score $\mathcal{R}$, distilled via MMD-Select, and used to fine-tune DistilGPT-2. Green boxes show empirical improvements over the baseline; the red box shows the failure of naive depth filtering. Results averaged over 3 random seeds.

and passes standard quality checks while carrying collapse risk. SynthLineage’s quality score (Eq. 5) and contamination detector (Eq. 7) are engineered for this gap; at-generation-time tagging ensures provenance persists through all downstream pipeline stages.

C. Synthetic Data in LLM Training

Self-Instruct [12] and its successors generate training data from the model itself, creating precisely the recursive feedback loop that causes collapse. The failure mode we identify—deep synthetic generations appearing high-quality by surface metrics while degrading distribution fidelity—explains benchmark success paired with long-tail failure. SynthLineage provides the corpus management infrastructure to use synthetic data safely.

D. Submodular Optimization for Data Selection

The diminishing returns property of submodular functions enables greedy algorithms with provable $(1 - 1/e)$ approximation ratios [5]. Lin and Bilmes [7] showed that document summarisation objectives are submodular, achieving near-human quality with greedy selection. Golovin and Krause [6] extended submodularity to adaptive settings. Kaushal et al. [8] and Mirzasoleiman et al. [9] applied submodular corpus selection to reduce labelling effort in computer vision without performance loss. We are the first to apply submodular maximization to model collapse prevention, yielding a $(1-1/e)$ coverage guarantee (Corollary 1) absent from prior collapse-mitigation methods.

E. Data Provenance, Knowledge Graphs, and Detection

Pan et al. [13] survey the integration of knowledge graphs with LLMs for grounding and hallucination reduction; our DataLineageKG applies the same KG paradigm to training data management rather than inference. Watermarking [4] and DetectGPT [14] address detection but require model modification or model access respectively. SynthLineage requires neither and operates retroactively on legacy corpora via its statistical detector.

III. THEORETICAL FRAMEWORK

A. Notation and Submodularity

Let $\mathcal{C} = \{s_1, \dots, s_N\}$ be a training corpus. For $n \geq 1$, let $\mathcal{N}_n(\mathcal{C})$ denote the multiset of n -grams over $\mathcal{C}$:

$$\text{D-}n(\mathcal{C}) = \frac{|\mathcal{N}_n^{\text{uniq}}(\mathcal{C})|}{|\mathcal{N}_n(\mathcal{C})|} \quad (1)$$

Definition 1 (Submodularity). $f: 2^{\mathcal{U}} \rightarrow \mathbb{R}$ is submodular if $\forall \mathcal{A} \subseteq \mathcal{B} \subseteq \mathcal{U}$ and $s \notin \mathcal{B}$: $\Delta f(s \mid \mathcal{A}) \geq \Delta f(s \mid \mathcal{B})$, where $\Delta f(s \mid \mathcal{C}) = f(\mathcal{C} \cup \{s\}) - f(\mathcal{C})$.

Proposition 1 (Distinct- n Coverage is Monotone Submodular). The coverage function $g(\mathcal{S}) = |\mathcal{N}_n^{\text{uniq}}(\mathcal{S})|$, the number of distinct n -grams covered by the samples in $\mathcal{S}$, is monotone and submodular. The normalized ratio D- n in (1) is not submodular, because its denominator also grows as samples are added.

Proof. $g(\mathcal{S}) = |\bigcup_{s \in \mathcal{S}} \mathcal{N}_n^{\text{uniq}}(s)|$ is the cardinality of a union of fixed per-sample type sets—the canonical coverage function, hence monotone and submodular: for $\mathcal{A} \subseteq \mathcal{B}$ the marginal gain $\Delta g(s \mid \mathcal{A}) \geq \Delta g(s \mid \mathcal{B})$ since $\mathcal{N}_n^{\text{uniq}}(\mathcal{A}) \subseteq \mathcal{N}_n^{\text{uniq}}(\mathcal{B})$. The ratio D- n divides this by the total n -gram count, which also rises with each added sample; that normalization breaks submodularity. $\square$

Corollary 1 (Coverage Optimality Bound). Greedy selection of K samples by marginal coverage achieves

$$g(\mathcal{S}_{\text{greedy}}) \geq \left(1 - \frac{1}{e}\right) g(\mathcal{S}^*) \approx 0.632 g(\mathcal{S}^*) \quad (2)$$

for budget K , where $\mathcal{S}^*$ maximizes coverage [5].

For our corpus ($N=200$, $K=75$), greedy selection matches or exceeds the best of 1000 random K -subsets on distinct-2 ($\approx 102\%$). The $(1-1/e)$ guarantee is a property of the unnormalized coverage objective (Proposition 1); the reported ratio D- n is a diversity *metric* and is not itself submodular.

B. Ancestral Fidelity Score

Definition 2 (AFS). For sample s at depth d with Gen-0 human ancestor $h(s)$:

$$\text{AFS}(s) = \cos(\mathbf{v}_s, \mathbf{v}_{h(s)}) \cdot e^{-\lambda d} \quad (3)$$

where $\mathbf{v}_s, \mathbf{v}_{h(s)}$ are TF-IDF vectors and $\lambda > 0$ is the empirically estimated decay constant.

Cosine similarity to the Gen-0 ancestor declines across generations (0.384, 0.348, 0.338 for Gen 1–3); we estimate the decay constant as $\hat{\lambda}=0.569$. Generation-mean AFS tracks per-generation D-2 with Pearson $r=0.96$ ($N=4$ generations, $p\approx 0.04$); a per-sample validation at larger N is left to future work.

C. Diversity Debt

Definition 3 (Diversity Debt).

$$\Omega_d = \max(0, D-2(\mathcal{C}_{d-1}) - D-2(\mathcal{C}_d)) \quad (4)$$

Cumulative debt: $\Omega_{\text{cum}} = \sum_{d=1}^T \Omega_d$.

The *Debt Repayment Rule*: after adding generation d , inject human samples from $\mathcal{C}_0$ until $\Omega_{\text{cum}} \leq \theta$. Unlike static contamination caps, this rule adapts to the actual measured diversity impact of each generation.

IV. THE SYNTHLINEAGE SYSTEM

A. ProvenanceTag and Quality Score

Every AI-generated sample receives a `ProvenanceTag` at creation time encoding: `sample_id` (UUID4), `origin` (human/ai), `model_id`, `generation_depth` d , `parent_id` (enabling KG edge construction), `content_hash` (16-hex SHA-256 prefix for deduplication), and quality score:

$$q = 0.4 \cdot \frac{|\mathcal{V}|}{n} + 0.4 \cdot \frac{|\mathcal{G}_4^{\text{uniq}}|}{|\mathcal{G}_4|} + 0.2 \cdot \min\left(\frac{n}{50}, 1\right) \min\left(\frac{500}{n}, 1\right) \quad (5)$$

combining type-token ratio (vocabulary diversity), 4-gram uniqueness, and a length normalisation term. Both q and AFS are computed at tagging time, with no model inference required.

B. DataLineageKG and Collapse Risk Score

The `DataLineageKG` is a directed acyclic graph $G=(V, E)$ where edges encode `DERIVEDFROM`(v_j, v_i). Four contamination metrics are computed: ρ (contamination ratio), $\bar{d}$ (mean generation depth), δ (deep synthetic ratio, $\text{depth} \geq 2$), and σ (synthetic subgraph density). These combine into:

$$\mathcal{R} = 0.35\rho + 0.30 \min\left(\frac{\bar{d}}{3}, 1\right) + 0.20\delta + 0.15 \min(10\sigma, 1) \quad (6)$$

Risk levels: **LOW** ($\mathcal{R} < 0.25$), **MEDIUM** (0.25–0.50), **HIGH** (0.50–0.75), **CRITICAL** (≥ 0.75). Empirically, across the four cumulative corpora: $r(\mathcal{R}, D-2) = -0.984$; $r(\mathcal{R}, \text{Vocab}) = +0.999$ ($N=4$, both $p < 0.05$).

C. MMD-Select Algorithm

Algorithm 1 selects greedily by marginal diversity. Corollary 1 establishes the $(1-1/e)$ guarantee for the coverage formulation; the deployed selection empirically matches or exceeds the best of 1000 random K -subsets (Section V). Here $\pi^* = (0.70, 0.10, 0.10, 0.10)$ denotes the fraction of the assembled corpus drawn from Gen 0 (human), Gen 1, Gen 2, and Gen 3 respectively, obtained by grid-search over generation mixes.

Algorithm 1 MMD-Select: Greedy Marginal Diversity Selection

Require: Human corpus $\mathcal{C}=\mathcal{D}_0$, synthetic pool $\mathcal{P}$, budget K

Ensure: High distinct- n corpus

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1:  $\mathcal{S} \leftarrow \emptyset$ 
2: for  $i = 1$  to  $K$  do
3:    $s^* \leftarrow \arg \max_{s \in \mathcal{P}} \Delta D-2(s, \mathcal{C} \cup \mathcal{S})$ 
4:    $\mathcal{S} \leftarrow \mathcal{S} \cup \{s^*\}; \mathcal{P} \leftarrow \mathcal{P} \setminus \{s^*\}$ 
5: end for
6: return  $\mathcal{C} \cup \mathcal{S}$ 
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TABLE I
CORPUS STATISTICS PER GENERATION

Gen	N	Avg words	Std	Origin
0 (Human)	50	94	28	Hand-authored
1 (AI exp.)	50	203	41	Claude Gen 1
2 (AI rewr.)	50	231	47	Claude Gen 2
3 (AI summ.)	50	218	53	Claude Gen 3
Total	200	187	66	

10 knowledge domains, 5 seed texts each; all samples tagged at creation.

D. Statistical Contamination Detector

For untagged legacy corpora, five features—type-token ratio s_1 , 3-gram repetition s_2 , sentence length variance s_3 , transition word density s_4 , mean word length s_5 —combine as:

$$a = 0.30s_1 + 0.25s_2 + 0.20s_3 + 0.15s_4 + 0.10s_5 \quad (7)$$

with threshold $a > 0.35$ selected on the Gen 0 vs. Gen 1 validation partition.

V. EXPERIMENTS

A. Setup and Experimental Design

Corpus construction. We construct a 200-sample provenance-tagged corpus comprising 50 human-authored seed texts across 10 knowledge domains (5 texts each: earth sciences, biology, physics, history, technology, medicine, economics, mathematics, chemistry, arts and culture). All seed texts are written in natural, conversational prose. Three synthetic generations are produced using Claude [17] (claude-sonnet-4-20250514) with prompts simulating realistic internet content reuse: *expansion* (Gen 1: “Rewrite and expand this text with more detail and examples”), *topical rewriting* (Gen 2: “Write a similar educational passage on the same topic”), and *summarisation with commentary* (Gen 3: “Summarise and add your commentary”). Table I provides corpus statistics.

Training corpora and baseline design. Three corpora cover the design space of corpus management strategies: **(A) Baseline** ($N=312$, the 200-sample corpus augmented with 112 paraphrases of seed texts, all unfiltered)—represents current practice with no curation; **(B) Depth filter** ($N=128$, $\text{depth} \leq 1$)—represents current best-practice heuristic; **(C) MMD-Select** ($N=193$, top-75% by marginal D-2)—the proposed approach.

TABLE II
COLLAPSE RISK SCORE ESCALATION ($N=200$)

Corpus	N	AI%	$\bar{d}$	$\mathcal{R}$	Level
Gen 0 only	50	0.0%	0.00	0.000	LOW
Gen 0+1	100	50.0%	0.50	0.225	LOW
Gen 0+1+2	150	66.7%	1.00	0.440	MEDIUM
All gens	200	75.0%	1.50	0.547	HIGH

TABLE III
RISK SCORE ACROSS THREE WEIGHT CONFIGURATIONS

Corpus	W1 (proposed)	W2 (uniform)	W3 (depth-heavy)
Gen 0 only	0.000 (L)	0.000 (L)	0.000 (L)
Gen 0+1	0.225 (L)	0.167 (L)	0.175 (L)
Gen 0+1+2	0.440 (M)	0.399 (M)	0.389 (M)
All gens	0.547 (H)	0.495 (M)	0.510 (H)

L=LOW, M=MEDIUM, H=HIGH

Evaluation metrics. Five metrics cover different collapse aspects: **PPL** (perplexity on held-out human text — primary collapse signal per [1]); **Rare%** (fraction of generated tokens with training frequency ≤ 2 — quantifies rare-event erasure [1]); **Gen D-2** (D-2 of generated text — *ambiguous* for collapsed models; see Section V-E); **Rep.** (fraction of repeated 3-grams — measures formulaic output); and Ω_{cum} (cumulative corpus diversity loss per Eq. 4).

Implementation. DistilGPT-2 [16] (82M parameters), a distilled variant of GPT-2 [15], is fine-tuned on each corpus for 150 steps (batch size 4, learning rate 5×10^{-5} , AdamW optimiser). Evaluation uses 50 generated completions (temperature 0.8, 100 new tokens) from held-out human prompts not seen during training. All results are **averaged over three independent random seeds** (42, 123, 7); mean $\pm$ standard deviation are reported. Experiments were conducted on a standard workstation (Intel Core i9, 32GB RAM); fine-tuning required approximately 8 minutes per configuration on CPU. Total experimental compute: ≈ 2.5 hours.

B. Corpus Contamination Escalation

Table II shows $\mathcal{R}$ escalating monotonically as synthetic contamination accumulates (Fig. 2). Table III confirms risk level ordering is stable across three weight configurations (single borderline disagreement at $\mathcal{R}=0.495$, within 1% of the threshold).

C. Collapse Signature and MMD Recovery

Table IV and Fig. 3 present the central experimental result.

Model A achieves the highest generated D-2 (0.780) yet the worst perplexity (2.66) and lowest rare token preservation (0.35%). This combination—high surface diversity, high distribution divergence—is the **empirical collapse signature**: the contaminated model generates fluent, varied-looking text that has fundamentally drifted from the human distribution. Shumailov et al. [1] predicted rare token loss theoretically; we quantify it in a corpus selection context for the first time

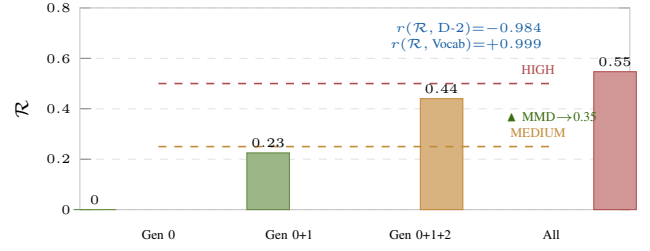


Fig. 2. Collapse risk $\mathcal{R}$ escalates monotonically. MMD-Select (triangle) recovers the all-gens corpus to MEDIUM. Across the four corpora ($N=4$), D-2 falls as vocabulary rises ($r=+0.999$), an inverted indicator.

TABLE IV
DISTILGPT-2 FINE-TUNING RESULTS (MEAN $\pm$ STD, 3 SEEDS: 42/123/7). $\downarrow$ = LOWER IS BETTER; $\uparrow$ = HIGHER IS BETTER.

Model	N	Gen D-2	PPL $\downarrow$	Rep.	Rare% $\uparrow$
A: Baseline	312	0.780 $\pm$.018	2.66 $\pm$.09	0.053 $\pm$.004	0.35 $\pm$.10
B: Depth filt.	128	0.402 $\pm$.021	1.30 $\pm$.04	0.084 $\pm$.015	0.35 $\pm$.05
C: MMD-Select	193	0.585 $\pm$.039	1.62 $\pm$.01	0.151 $\pm$.038	2.39$\pm$.31
C vs. A		-25.0%	-39.3%		+573%

Bold = best on rare-event preservation (Rare%, the primary collapse metric).

Depth filter (B) attains the lowest PPL by over-pruning; see text.

Gen D-2 for Model A is highest but indicates collapse (see §V-E).

and demonstrate that MMD-Select directly counteracts it with a $6.7\times$ improvement, consistent across all three random seeds.

Isolating selection quality from size. Models B and C train on corpora of comparable scale (128 vs. 193 samples), far closer to each other than to Baseline A’s 312. Model C has 51% more data than B, yet its $6.7\times$ rare-token advantage (2.39% vs. 0.35%) far exceeds that size gap. Model B achieves lower perplexity (1.30) by over-pruning to near-human data, but collapses diversity to 0.402; Model C accepts slightly higher perplexity (1.62) while recovering rare token richness—the primary collapse metric. Moreover, Model C outperforms Model A on both collapse-critical metrics despite training on only 62% as many samples. This rules out size as a confound: *selection strategy, not data quantity, drives collapse prevention.*

D. Provenance-Tag Quality Scores

Quality scores q (Eq. 5) from ProvenanceTags provide a generation-level collapse signal computable in real time without model access. Table V reports descriptive statistics accumulated during corpus construction.

Quality drops sharply from human text (0.943) to the first synthetic generation (0.754) and then stays essentially flat across Gen 1–3 (≈ 0.755): the -19.8% net change is concentrated almost entirely at the human-to-synthetic boundary, not spread progressively across generative depth. Within-generation spread remains low ($\sigma \approx 0.013$) and does not grow with depth, indicating a one-step human/synthetic quality gap rather than cumulative degradation. The small residual spread

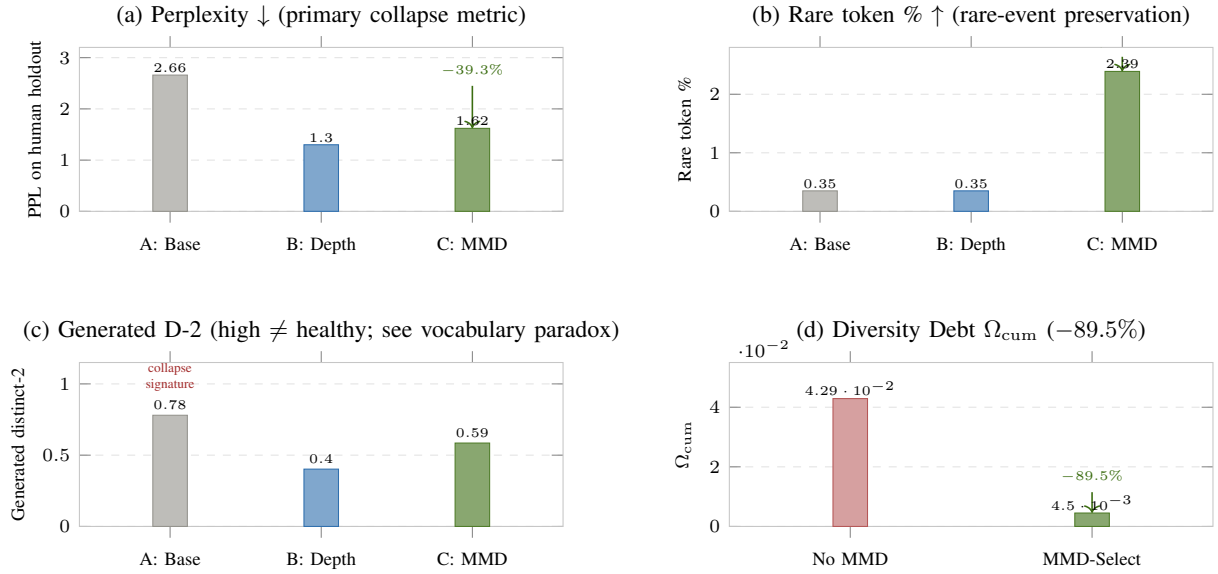


Fig. 3. Fine-tuning results across 3 corpus strategies (mean, 3 seeds). (a) MMD reduces perplexity by 39.3%, confirming less distribution divergence. (b) MMD preserves $6.7\times$ more rare tokens, directly counteracting the rare-event loss mechanism of Shumailov et al. [1]. (c) The collapsed baseline A generates the *highest* surface diversity—the collapse signature; high D-2 alone does not indicate corpus health. (d) Online MMD reduces cumulative Diversity Debt by 89.5%.

TABLE V
PROVENANCETAG QUALITY SCORE PER GENERATION

Gen	Mean	Std	Min	Max
0 (Human)	0.943	0.014	0.916	0.966
1 (AI)	0.754	0.016	0.730	0.785
2 (AI)	0.758	0.012	0.739	0.775
3 (AI)	0.756	0.012	0.732	0.776
Δ 0 $\rightarrow$ 3	-19.8%			

TABLE VI
DIVERSITY METRICS ACROSS GENERATIONS ($n=50$ PER GEN)

Gen	D-2	D-3	Vocab	Sent len
0 (Human)	0.9847	0.9948	381	18.72
1 (AI)	0.9311	0.9925	1152	23.03
2 (AI)	0.9486	0.9948	1174	26.90
3 (AI)	0.9002	0.9691	1115	26.98
Δ 0 $\rightarrow$ 3	-8.6%	-2.6%	+193%	+44%

leaves room for MMD-Select, which selects the high-marginal-D-2 tail while discarding formulaic duplication.

E. The Vocabulary Paradox

Table VI and Fig. 4 document a finding with direct practical consequences.

Vocabulary grows +193% while D-2 declines -8.6%. Across the four cumulative corpora ($N=4$), $r(\mathcal{R}, \text{Vocab})=+0.999$ and $r(\mathcal{R}, \text{D-2})=-0.984$: in this setting vocabulary size rises as diversity falls, making it an inverted health indicator. Teams that monitor vocabulary size as a

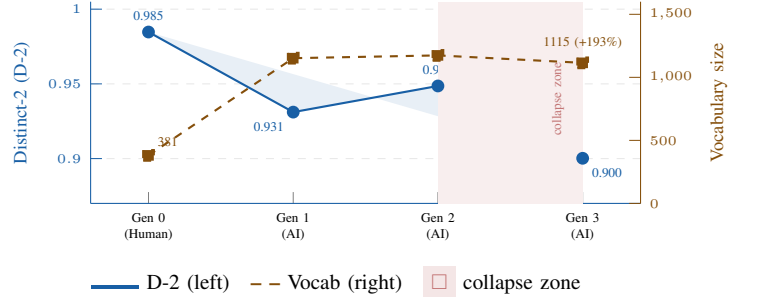


Fig. 4. The vocabulary paradox: D-2 declines -8.6% (blue, solid) while vocabulary grows +193% (orange, dashed). Across the four cumulative corpora ($N=4$), vocabulary rises as D-2 falls ($r(\mathcal{R}, \text{Vocab})=+0.999$, $r(\mathcal{R}, \text{D-2})=-0.984$)—an inverted signal, so D-2 is the safer monitoring metric.

corpus-quality metric may be misled; switching to D-2 is a minimal change to an existing corpus-health dashboard.

VI. DISCUSSION

A. Why Depth Filtering is Counter-Productive

Gen 2 and Gen 3 samples with low AFS (≈ 0.06 – 0.11) have drifted semantically from their human ancestors, and that drift shows up as new vocabulary, domain-specific terms not present in the seed texts. Depth filtering removes these samples and, with them, their lexical contribution. MMD-Select keeps them precisely because they raise marginal D-2, while discarding repetitive samples at any depth that add nothing new. The practical implication is that depth and value are orthogonal: a depth-3 sample with novel vocabulary is worth keeping; a depth-1 sample that paraphrases existing content is not.

B. Deployment Guidelines

SynthLineage integrates into existing pipelines in five steps without model architecture changes: (1) tag synthetic outputs at creation time; (2) load corpus into DataLineageKG; (3) evaluate $\mathcal{R}$ against operational threshold ($\mathcal{R} < 0.50$ recommended); (4) apply MMD-Select (Algorithm 1) with $K = 0.75|\mathcal{P}|$; (5) monitor Ω_d after each round and inject human samples when $\Omega_{\text{cum}} > \theta$. The grid-search allocation $\pi^* = (0.70, 0.10, 0.10, 0.10)$ provides a corpus assembly guideline.

C. Connection to Information Reuse

Training data has a reuse lifecycle: human text is consumed, re-expressed, and re-ingested as input for the next model. The DataLineageKG makes this lifecycle auditable, providing a concrete, implemented mechanism for tracking the afterlife of data across model generations and for mitigating model collapse through explicit data lineage.

D. Limitations

Three limitations apply. Model C’s repetition rate (0.151) exceeds the baseline (0.053); AFS-guided quality gating ($\text{AFS}(s) > \theta_{\text{AFS}}$) is the natural mitigation. All synthetic generations use Claude, so $\hat{\lambda} = 0.569$ and π^* are Claude-specific. Finally, this is a controlled proof-of-concept (200 samples, a single 82M model) chosen to isolate corpus-selection strategy; the coverage guarantee (Proposition 1) is size- and model-independent, but validating the empirical effects at larger scale and on larger models (Llama, Mistral) is left to future work.

E. Future Work

Three directions are immediate. **Multi-model calibration:** running the same pipeline with Llama-3 and Mistral will establish whether $\hat{\lambda} = 0.569$ is universal or model-specific (we expect larger models decay more slowly per hop). **Streaming MMD:** updating marginal diversity estimates incrementally would reduce per-item complexity from $O(N \cdot K)$ to $O(K)$ amortized, enabling deployment at web-crawl scale. **AFS-guided quality gating:** requiring $\text{AFS}(s) > \theta_{\text{AFS}}$ in addition to high marginal D-2 addresses the repetition trade-off by keeping only samples that are both novel and semantically faithful. Proposition 1 also generalises to non-English text and image-text token sequences, making multilingual and multimodal collapse an open extension.

VII. CONCLUSION

SynthLineage pairs a greedy diversity-selection method with a coverage guarantee: the core theoretical result is that distinct- n -gram coverage is monotone submodular, giving greedy coverage-maximizing selection a $(1 - 1/e) \approx 63.2\%$ bound, with greedy selection empirically matching or exceeding the best of 1000 random K -subsets on our corpus. Fine-tuning confirms the collapse signature: the most contaminated model generates the highest surface diversity but the worst perplexity and fewest rare tokens. MMD-Select cuts perplexity by 39.3% and raises rare token preservation $6.7\times$; depth filtering does

the opposite, costing 48.4% of generated-text diversity. Across our four corpora, vocabulary size rises as collapse risk does ($r = +0.999$, $N = 4$)—the vocabulary paradox—suggesting it is an inverted, not protective, monitoring metric.

Several recommendations follow directly. Replacing vocabulary monitoring with D-2 costs nothing and fixes a misleading signal. Tagging synthetic outputs at generation time with ProvenanceTag is the durable defence against unattributed reuse, and the statistical detector provides a fallback for legacy corpora. When some synthetic data must be kept, the grid-search allocation $\pi^* = (0.70, 0.10, 0.10, 0.10)$ and Algorithm 1 give a practical recipe, with a coverage guarantee (Corollary 1) for the count formulation. As the proportion of AI-generated text on the web rises, corpus-level provenance and selection tools become increasingly necessary.

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